

JY Liu

Solutions Architect | Technical Lead

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Summary

- Solutions architect and technical lead with 12+ years shipping web, mobile, cloud, and integration-heavy platforms across fintech, SaaS, banking, consular, and enterprise environments.
- Hands-on across frontend, backend, APIs, DevOps, and cloud infrastructure — consistently taking products from discovery and architecture through production delivery.
- Leverages AI tooling pragmatically across the SDLC — analysis, implementation, prototyping, and quality assurance — while building AI-native products end-to-end and maintaining production-grade technical rigor.
- All engagements delivered as independent consulting or contracting work; overlapping dates reflect concurrent client engagements.

Experience

Vailley AI Solution Ptd Limited, Senior Web Developer

Singapore

- Leading cross-functional delivery for **VFS Global's** visa and consular platforms, combining solution architecture with hands-on implementation across application, integration, and data workflows.
- Delivered a Consular Dashboard with RBAC and ABAC controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for agents, and SAP Concur to OneVasco integration for **VFS Global**.
- Built a parallelized multi-file **PDF scanner** microservice for consular document processing, with pre-warmed S3 clients, deterministic file ordering, and dedicated e2e test coverage.

Jun 2025 – Present

1 year 3 months

Shift Markets, Senior Software Engineer

Remote

- Lead engineer for **CXM Direct** (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution with TypeScript, React, and Next.js.
- Engineered a **Unified Payment Gateway** integrating 20+ global Payment Service Providers (PSPs), unifying fragmented payment workflows into a single scalable infrastructure.
- Strengthened system reliability and delivery velocity through architectural cleanup, performance tuning, and production-oriented engineering practices.

Jan 2023 – Present

3 years 8 months

InternetBrands, Senior Web Engineer

Malaysia

- Maintained and improved the **Dentrix Hub Forms** (Online Form for Dentrix & HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scalability.
- Introduced PHP linting and unit testing standards, increasing code coverage across the code-base.
- Managed Kubernetes deployment with Helm and Rancher for scalable, reliable infrastructure.

Apr 2023 – Sept 2024

1 year 6 months

MaiDaLv, Senior Full Stack Developer

Remote

- Built Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subscriptions, WeChat integration, and IP-risk analysis.

Feb 2025 – Apr 2025

2 months

Your Software Supplier, Full Stack Developer

Remote

- Delivered full-stack web solutions for client engagements, concurrent with InternetBrands role.

Aug 2023 – Jan 2024

6 months

Cake Experiential Communications Asia, Solutions Architect

Kuala Lumpur, Malaysia

- Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.
- Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems.

Apr 2021 – May 2023

2 years 2 months

Glints , Web Software Engineer - Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design. - Strengthened dashboard responsiveness and release quality through improved testing and deployment automation.	Remote Jan 2022 – Dec 2022 1 year
InnoLab Global , Senior Information Technology Consultant - Delivered cross-platform products across Angular, Ionic, Flutter, Firebase, and AWS while automating cloud delivery workflows and mentoring junior developers. - Shipped the Forest City Super App production MVP (90% complete) on a Turborepo + Next.js + Prisma 7 + PostgreSQL 18 + Auth.js v5 stack covering community, events, payments, partner portal, admin console, and mini-app runtime. - Built Aegis Shooting Academy (ASA) digital platform end-to-end with NestJS + Prisma back-end, Next.js admin dashboard, and React Native/Expo mobile application. - Delivered a corporate website and WooCommerce product catalogue for 3V Malaysia (furniture manufacturer) using Docker Compose with a premium Apple-inspired theme direction. - Architected and led delivery for AvaEra's MLM compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js 16, Prisma 7, PostgreSQL 18, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, and merchant profit share. - Delivered AvaEra sub-products including the DaiseE Beauty Series compensation plan amendment and DaiseE Leader inventory credit + redemption lifecycle, with end-to-end Playwright e2e coverage. - Built and shipped HappyAI (MVP COMPLETE v1.0.0) for Gembira Bina Sdn Bhd, an AI-powered tender BQ extraction, validation, and T1121 xlsx generation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.	Kuala Lumpur, Malaysia Sept 2018 – Present 8 years
Supahands , Full Stack Developer Contributed to full-stack product development on Supahands' distributed-team platform.	Remote Feb 2018 – Jun 2018 5 months
The Consociation LLP , Information Technology Consultant Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.	Singapore Sept 2016 – Apr 2020 3 years 8 months
Buddy Holdings Pte Ltd , Senior Information Technology Consultant - Architected and led delivery of the Buddy System Asia MLM operations platform — Angular + Laravel + MySQL + Kafka covering hierarchy management, complex point calculations, point redemption, inventory tracking, and family tree updates at scale. - Built intuitive agent-facing UI for tracking progress, points, and family tree structures in real time.	Malaysia Sept 2016 – Jan 2024 7 years 5 months
Fuji Xerox , Embedded Software Engineer Developed embedded software for printer/scanner hardware platforms with C/C++.	Malaysia Feb 2016 – May 2016 4 months
Netccentric , Software Developer Contributed to full-stack web product development across digital marketing and advertising workflows.	Malaysia Oct 2015 – Dec 2015 3 months
Nexstream , Hybrid App Developer Built hybrid mobile applications bridging web technologies and native iOS/Android platforms.	Malaysia Jun 2014 – Jul 2015 1 year 2 months

Selected Projects

Unified Payment Gateway Specialized payment infrastructure unifying 20+ global Payment Service Providers (PSPs) into a single, high-concurrency API layer. • Architected and delivered a unified integration layer for diverse PSPs, handling complex transaction workflows, reconciliation, and automated settlement. • Expanded regional coverage in 2026 with OmPay (IDR qris/va/dana, MYR duitnow per FpxAPI \$5.1/\$8.0), NexusPay chain callbacks, and PH bank-code refresh workflows. • Secured financial operations through strict audit logging, sensitive data isolation, and robust error-recovery mechanisms.	Jan 2023 – Present
VFS Global Consular Platform Mission-critical visa and consular services platform delivering digital transformation for global diplomatic missions. • Delivered a high-security Consular Dashboard with granular RBAC/ABAC and integrated SAP Concur for automated expense and travel management. • Built a digital corporate visa platform enabling agents to manage large-scale applications with real-time tracking and ETL-backed analytics.	Jun 2024 – Present
Superpowered Talent platform delivered with scalable web architecture, backend APIs, GraphQL, PostgreSQL, and event-driven workflows. • Built a responsive Next.js and Koa.js application with GraphQL integration, backend optimization, and scalable recruiter-facing workflows.	Jan 2022 – Dec 2022
Dentrix Hub Forms (Online Form for Dentrix & HOP) Dental practice patient form management platform — Laravel backend with Kubernetes deployment, PHP linting, and unit testing standards. • Maintained and improved the Laravel backend, strengthening response times and concurrency handling for patient form workflows. • Introduced PHP linting and unit testing, increasing code coverage across the codebase. • Managed Kubernetes deployment with Helm and Rancher for scalable, reliable infrastructure.	Apr 2023 – Sept 2024

Education

Master of University of Malaya, Business Analytics Business Administration	Kuala Lumpur, Malaysia Oct 2023 – Oct 2026
Master of University of Malaya, Mobile Cloud Computing Computer Science	Kuala Lumpur, Malaysia Jan 2011 – Jan 2013
Bachelor of Universiti Sains Malaysia, Software Engineering Engineering	Malaysia Jan 2008 – Jan 2011

Skills

Full Stack Engineering: TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic
Architecture, Cloud & DevOps: Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery
Backend, Data & Integration: PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems
AI-Enabled Delivery & Multi-Agent Systems: Expert at orchestrating LLM-based agents, RAG (Retrieval-Augmented Generation), prompt engineering, and custom GPT development to accelerate product delivery while maintaining technical rigor.

Languages

- English — Professional
- Malay — Professional
- Mandarin — Native
- Cantonese — Professional
- Japanese — Beginner
- Korean — Beginner